Green's Theorem (short lecture)

Thursday, May 29, 2025 2:55 PM

15 – Green's Theorem in the plane

15.1

THEOREM 1

Green's Theorem in the Plane[4]
(Transformation between Double Integrals and Line Integrals)

Let R be a closed bounded region (see Sec. 10.3) *in the xy-plane whose boundary C consists of finitely many smooth curves* (see Sec. 10.1). *Let $F_1(x, y)$ and $F_2(x, y)$ be functions that are continuous and have continuous partial derivatives $\partial F_1/\partial y$ and $\partial F_2/\partial x$ everywhere in some domain containing R. Then*

$$\iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) dx\, dy = \oint_C (F_1\, dx + F_2\, dy). \tag{1}$$

Here we integrate along the entire boundary C of R in such a sense that R is on the left as we advance in the direction of integration (see Fig. 234).

15.2 Recall:

When $\vec{F} = F_1(x, y)\hat{\imath} + F_2(x, y)\hat{\jmath}$

Use embedding to write

$$\vec{F} = F_1(x,y)\hat{\imath} + F_2(x,y)\hat{\jmath} + 0\hat{k}$$

and get

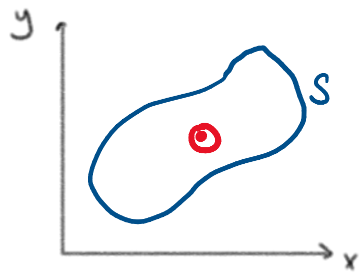
$$\vec{\nabla} \times \vec{F} = \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)\hat{k}$$

Hence Green's Theorem is

$$\iint_R (\operatorname{curl} \mathbf{F}) \cdot \mathbf{k}\, dx\, dy = \oint_C \mathbf{F} \cdot d\mathbf{r}.$$

Note that for a surface S on the xy-plane:

$\hat{k}$ is the direction



$\hat{k}$ is the direction of a unit normal to the surface

Later, Green's Theorem will generalize to Stokes' Theorem:

$$\iint_S \vec{F}\cdot\hat{n}\,dS = \oint_{\partial S} \vec{F}\cdot d\vec{r}$$

15.3

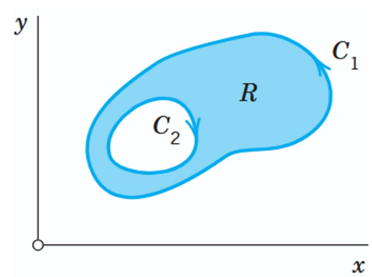
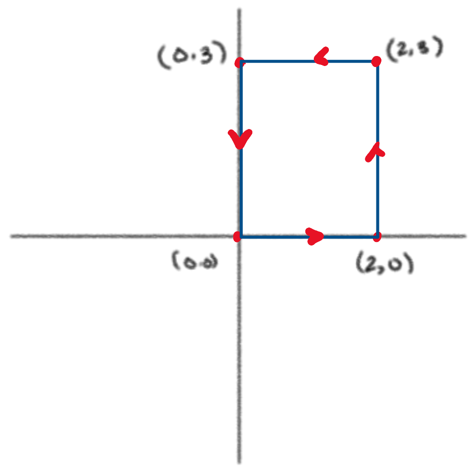


Fig. 234. Region R whose boundary C consists of two parts: C_1 is traversed counterclockwise, while C_2 is traversed clockwise in such a way that R is on the left for both curves

15.4 Example: Evaluate $\int_C \vec{F}(\vec{r})\cdot d\vec{r}$, where C is the counterclockwise boundary of the region R. using Green's Theorem, where

$\vec{F} = [x^2e^y, y^2e^x]$. R is the rectangle:



Note that we can describe the region R as

$$R = \{(x,y) : 0 \le x \le 2,\ 0 \le y \le 3\}$$

So then by Green's Theorem:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial x} \right) dx\,dy$$

$$= \int_0^3 \int_0^2 \left(\frac{\partial}{\partial x}(y^2 e^x) - \frac{\partial}{\partial y}(x^2 e^y) \right) dx\,dy$$

$$= \int_0^3 \int_0^2 \left[e^x y^2 - e^y x^2 \right] dx\,dy$$

$$= \int_0^3 \left[e^x y^2 - \frac{e^y x^3}{3} \right]_0^2 dy$$

$$= \int_0^3 \left[(e^2 - 1) y^2 - e^y \left(\frac{8}{3}\right) \right] dy$$

$$= \left[(e^2 - 1)\frac{y^3}{3} - \frac{8}{3} e^y \right]_0^3$$

$$= \left[(e^2 - 1)\frac{27}{3} - \frac{8}{3}(e^3 - 1) \right.$$

$$= 9(e^2 - 1) - \frac{8}{3} e^3 + \frac{8}{3}$$

$$= 9e^2 - \frac{8}{3} e^3 - 9 + \frac{8}{3}$$

$$= 9e^2 - \frac{8}{3}e^3 - 9 + \frac{8}{3}$$

$$= 9e^2 - \frac{8}{3}e^3 - \frac{19}{3} \approx 6.6067$$

15.5

Area of a Plane Region as a Line Integral Over the Boundary

In (1) we first choose $F_1 = 0$, $F_2 = x$ and then $F_1 = -y$, $F_2 = 0$. This gives

$$\iint_R dx\, dy = \oint_C x\, dy \qquad \text{and} \qquad \iint_R dx\, dy = -\oint_C y\, dx$$

respectively. The double integral is the area A of R. By addition we have

$$A = \frac{1}{2}\oint_C (x\, dy - y\, dx) \tag{4}$$

where we integrate as indicated in Green's theorem. This interesting formula expresses the area of R in terms of a line integral over the boundary. It is used, for instance, in the theory of certain **planimeters** (mechanical instruments for measuring area). See also Prob. 11.

For an ***ellipse*** $x^2/a^2 + y^2/b^2 = 1$ or $x = a \cos t$, $y = b \sin t$ we get $x' = -a \sin t$, $y' = b \cos t$; thus from (4) we obtain the familiar formula for the area of the region bounded by an ellipse,

$$A = \frac{1}{2}\int_0^{2\pi} (xy' - yx')\, dt = \frac{1}{2}\int_0^{2\pi} [ab \cos^2 t - (-ab \sin^2 t)]\, dt = \pi ab.$$